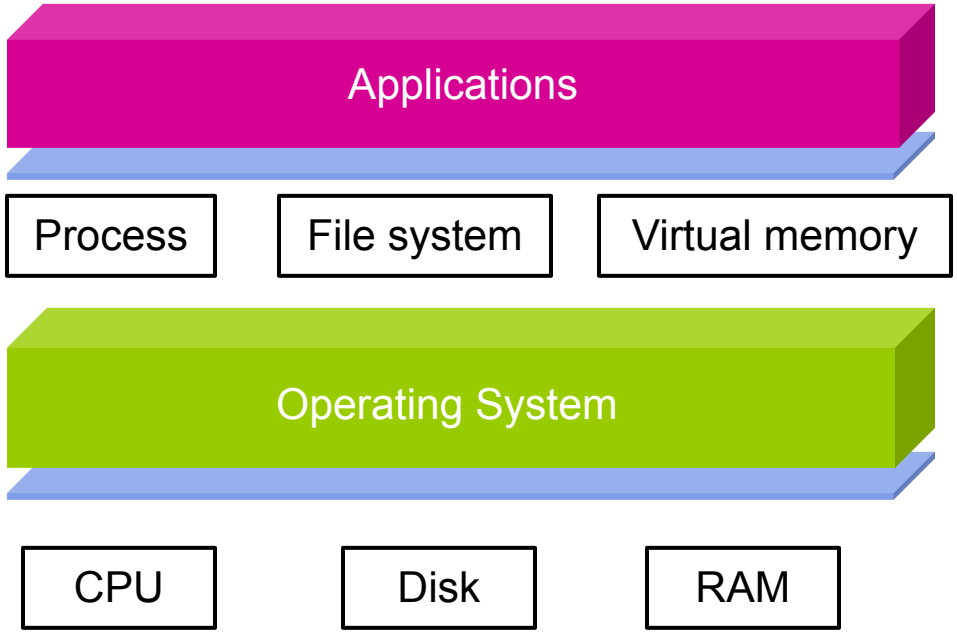
Processes and Threads

CS 202: Advanced Operating Systems

OS Abstractions



Today, we start discussing the first abstraction that enables us to virtualize (i.e., share) the CPU – processes!

Why we need the Process abstraction

- Imagine opening two calculator apps at the same time
  - Same program, each runs independently
- You can. Enter different numbers in each
  - No interference
- How? The process abstraction!
  - Each process has its own memory, state, and resources
  - This isolation ensures safe multitasking

Program

- Program is a file describing a computation
  - Executable code (machine instructions)
  - Data (info manipulated by the instructions

Process

- The process is the OS abstraction for execution
  - It is the unit of execution
  - It is the unit of scheduling

Process

- The process is the OS abstraction for execution
  - It is the unit of execution
  - It is the unit of scheduling
- A process is a program in execution
  - Programs are static entities with the potential for execution
  - Process is the dynamic/active entity of a program
    - Includes dynamic state
    - As the representative of the running program, it is the “owner” of other resources (memory, files, sockets, …)

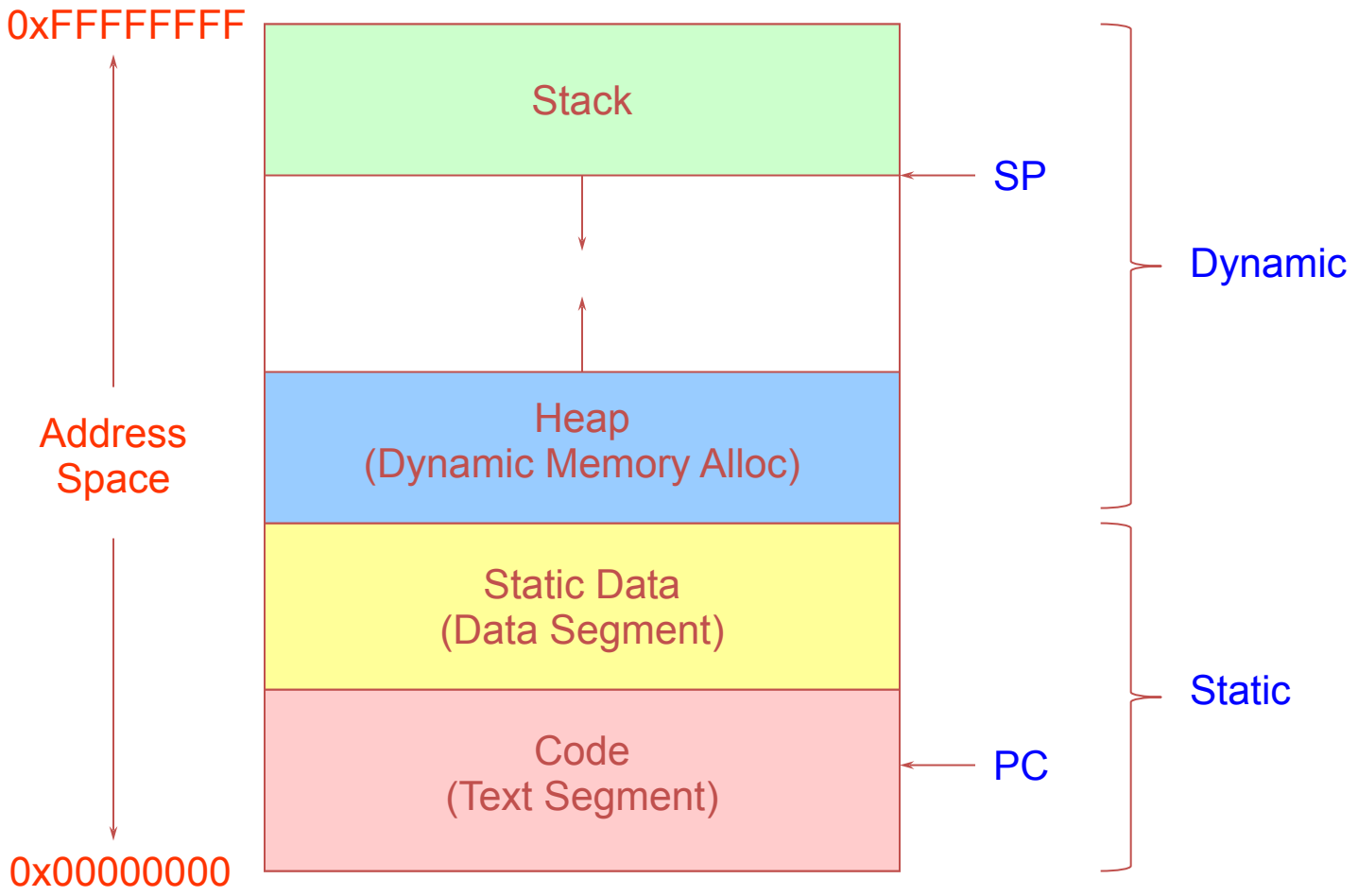
Process = Program ???

- A program is passive
  - Code + data
- A process is alive:
  - Code + data + stack +registers + PC…
- Same program can be run simultaneously
  - 2 processes

Process Components

- A process contains all the state for a program in execution
  - An address space containing
    - Static memory:
      - The code and input data for the executing program
    - Dynamic memory:
      - The memory allocated by the executing program
      - An execution stack encapsulating the state of procedure calls
  - Control registers such as the program counter (PC)
  - A set of general-purpose registers with current values
  - A set of operating system resources
    - Open files, network connections, etc.
- A process is named using its process ID (PID)

Address Space (memory abstraction)



Process Execution State

- A process is born, executes for a while, and then dies
- The process execution state that indicates what it is currently doing
  - Running: Executing instructions on the CPU
    - It is the process that has control of the CPU
    - How many processes can be in the running state simultaneously?
  - Ready: Waiting to be assigned to the CPU
    - Ready to execute, but another process is executing on the CPU
  - Waiting: Waiting for an event, e.g., I/O completion
    - It cannot make progress until event is signaled (disk completes)

Execution state (cont'd)

- As a process executes, it moves from state to state
  - Unix: "ps -x": STAT column indicates execution state
  - What state do you think a process is in most of the time?
  - How many processes can a system support?

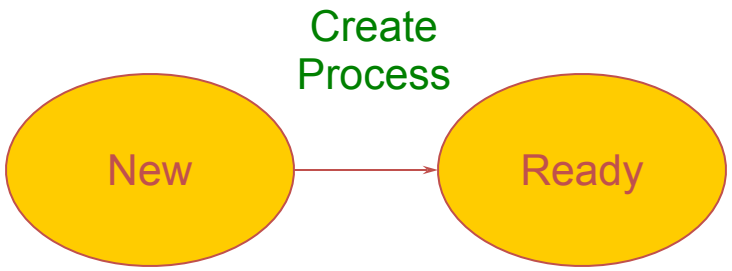
```
PROCESS STATE CODES
       Here are the different values that the s, stat and state output specifiers (header "S
       D    uninterruptible sleep (usually IO)
       R    running or runnable (on run queue)
       S    interruptible sleep (waiting for an event to complete)
       T    stopped, either by a job control signal or because it is being traced.
       W    paging (not valid since the 2.6.xx kernel)
       X    dead (should never be seen)
       Z    defunct ("zombie") process, terminated but not reaped by its parent.

       For BSD formats and when the stat keyword is used, additional characters may be displ
       <    high-priority (not nice to other users)
       N    low-priority (nice to other users)
       L    has pages locked into memory (for real-time and custom IO)
       s    is a session leader
       l    is multi-threaded (using CLONE_THREAD, like NPTL pthreads do)
       +    is in the foreground process group.
```

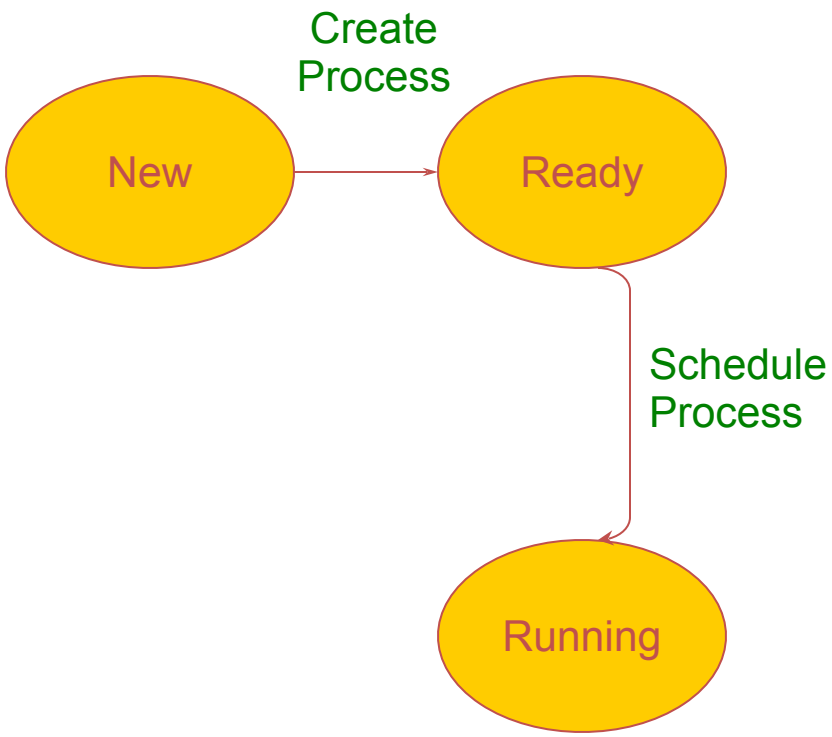
Execution State Graph



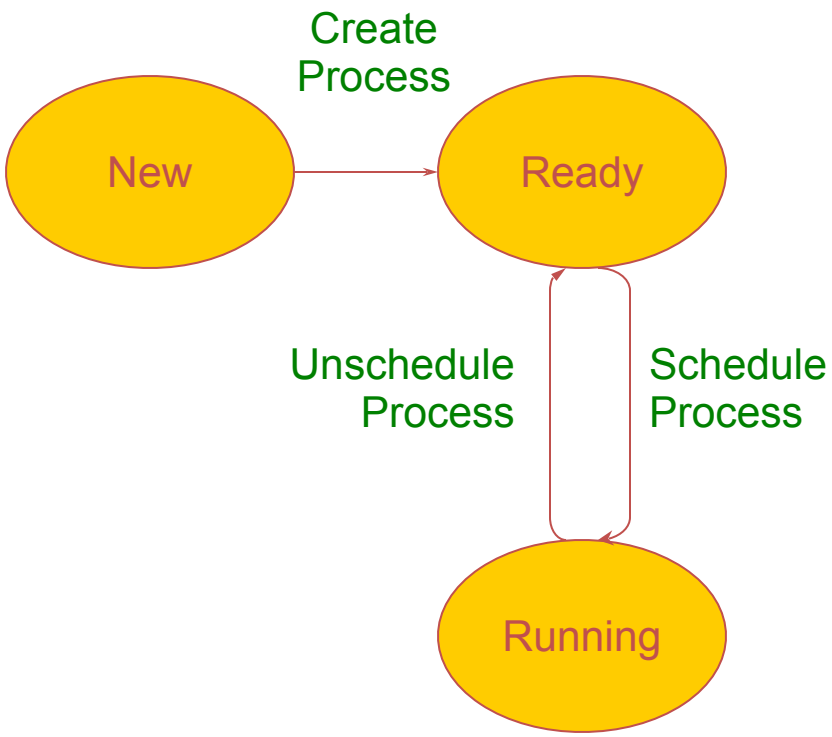
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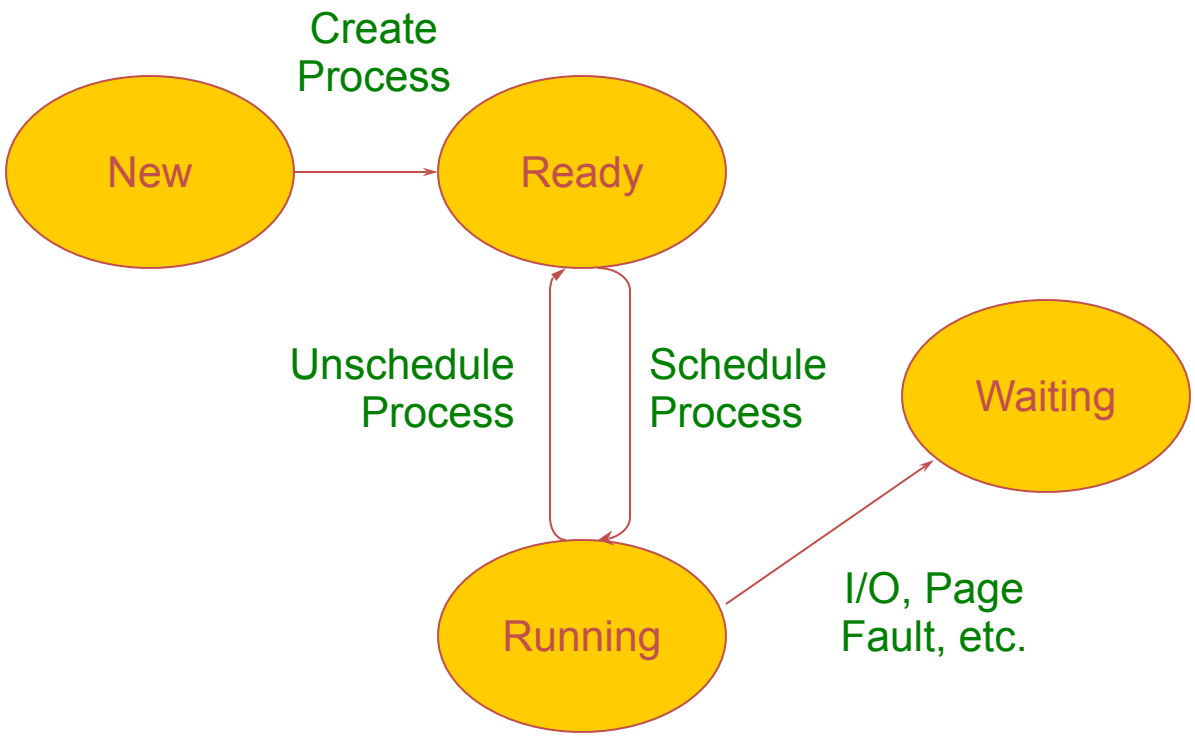
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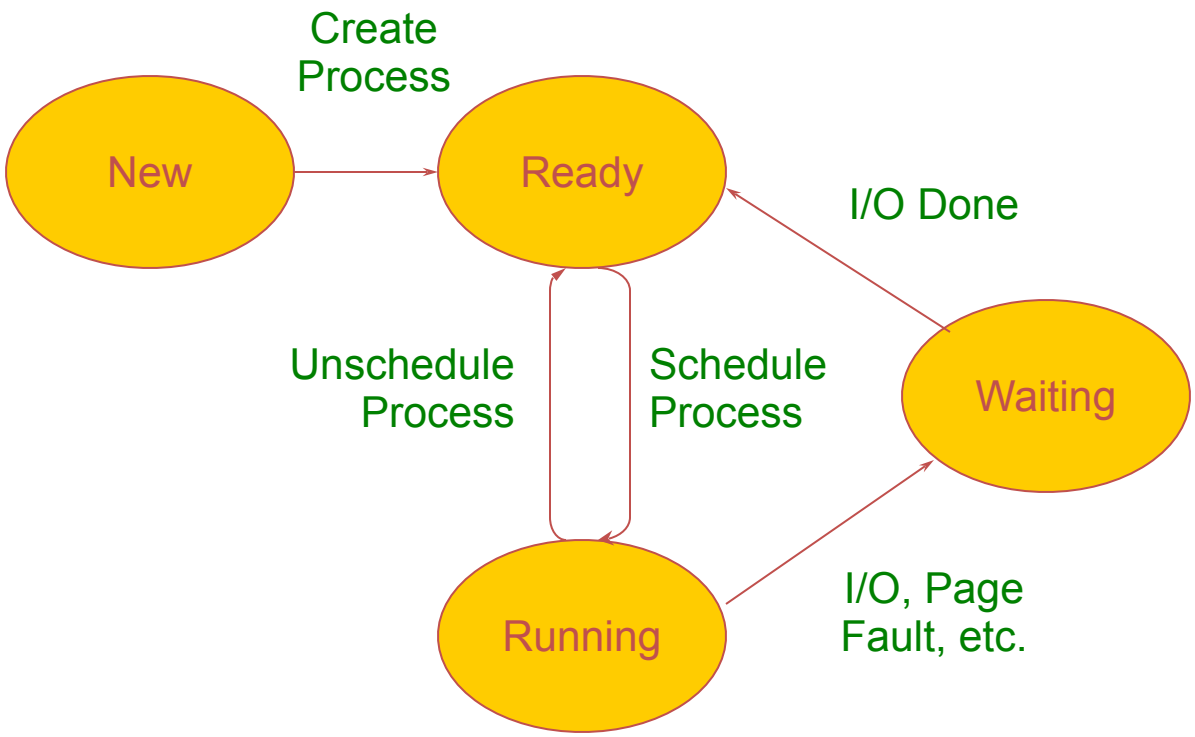
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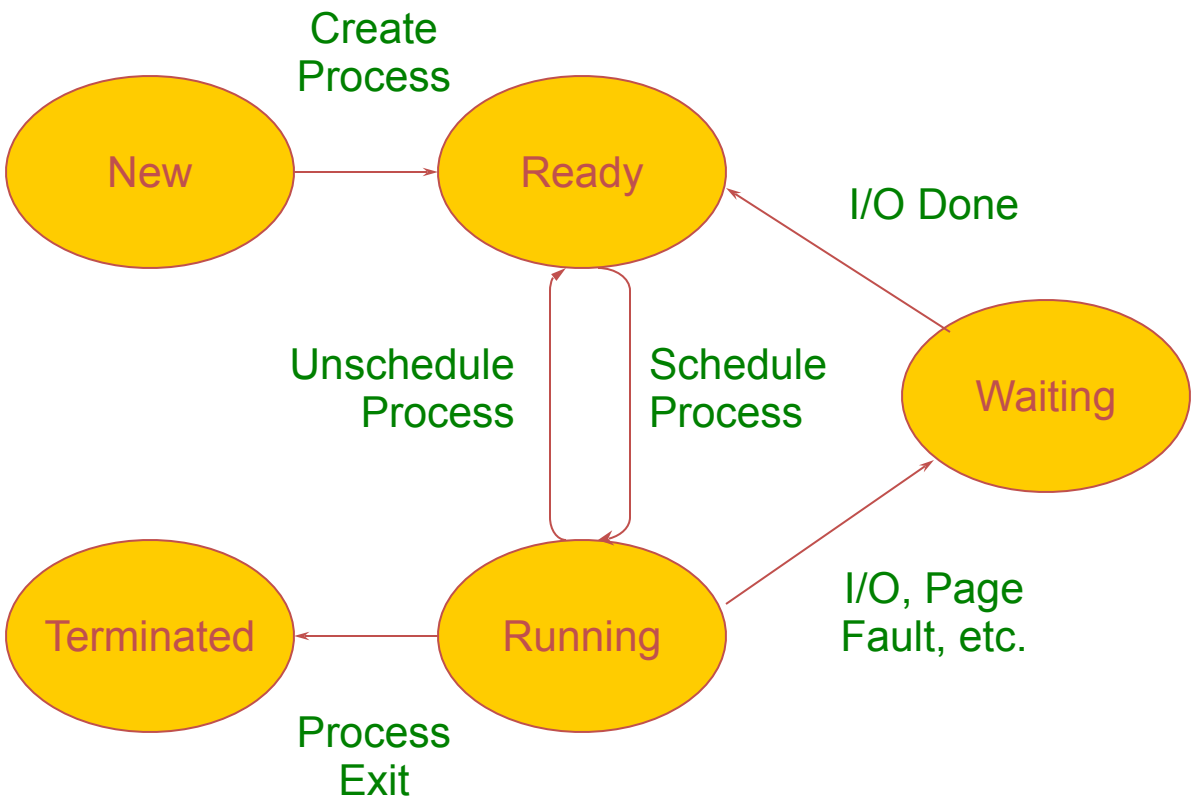
Execution State Graph



Execution State Graph



Execution State Graph



How does the OS support this model?

We will discuss three issues:

1. How does the OS represent a process in the kernel?
   - The OS data structure representing each process is called the Process Control Block (PCB)

2. How do we pause and restart processes?
   - We must be able to *save* and *restore* the full machine state

3. How do we keep track of all the processes in the system?
   - A lot of queues!

PCB Data Structure

- PCB is also where OS keeps all of a process' hardware execution state when the process is not running
  - Process ID (PID)
  - Execution state
  - Hardware state: PC, SP, regs
  - Location in memory
  - Scheduling info
  - User info
  - Pointers for state queues
  - Etc.
- This state is everything that is needed to restore the hardware to the same configuration it was in when the process was switched out of the hardware

xv6/proc.h: struct proc

```
00027 enum procstate { UNUSED, EMBRYO, SLEEPING, RUNNABLE, RUNNING, ZOMBIE };
00028
00029 // Per-process state
00030 struct proc {
00031   char *mem;                   // Start of process memory (kernel address)
00032   uint sz;                     // Size of process memory (bytes)
00033   char *kstack;                // Bottom of kernel stack for this process
00034   enum procstate state;        // Process state
00035   volatile int pid;            // Process ID
00036   struct proc *parent;         // Parent process
00037   struct trapframe *tf;        // Trap frame for current syscall
00038   struct context *context;     // Switch here to run process
00039   void *chan;                  // If non-zero, sleeping on chan
00040   int killed;                  // If non-zero, have been killed
00041   struct file *ofile[NOFILE];  // Open files
00042   struct inode *cwd;           // Current directory
00043   char name[16];               // Process name (debugging)
00044 };
```

How to pause/resume a process?

- When a process is running, its dynamic state is in memory and some hardware registers
  - Hardware registers include program counter, stack pointer, control registers, data registers, …
  - To stop and restart a process, we need to completely restore this state

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- When the OS stops running a process, it saves the current values of the registers (usually in PCB)
- When the OS resumes a process, it loads the hardware registers from the stored values in PCB

How to pause/resume a process?

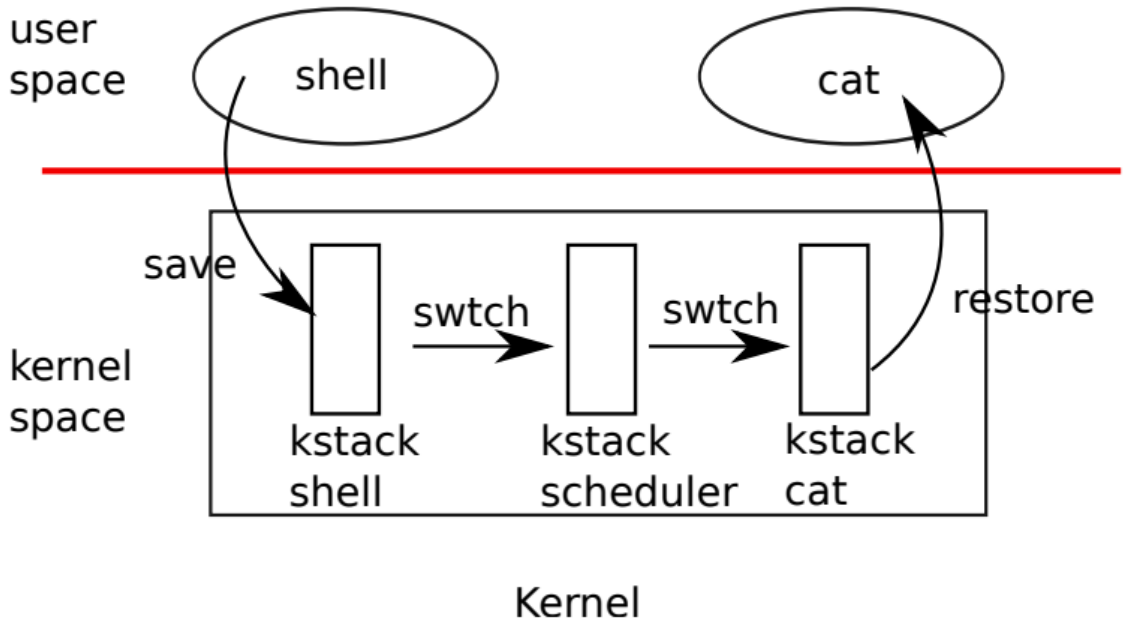
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- When the OS stops running a process, it saves the current values of the registers (usually in PCB)
- When the OS resumes a process, it loads the hardware registers from the stored values in PCB

- Changing CPU hardware state from one process to another is called a context switch
  - This can happen 100s or 1000s of times a second!

xv6: context switching

- Switching from one user process to another (shell to cat)
  - First moves to the kernel space, then calls swtch() for context switching



- In xv6, each CPU has a scheduler thread (scheduler() in proc.c)

xv6: PCB and context

```
00027 enum procstate { UNUSED, EMBRYO, SLEEPING, RUNNABLE, RUNNING, ZOMBIE };
00028
00029 // Per-process state
00030 struct proc {
00031   char *mem;
00032   uint sz;
00033   char *kstack;
00034   enum procstate state;
00035   volatile int pid;
00036   struct proc *parent;
00037   struct trapframe *tf;
00038   struct context *context;
00039   void *chan;
00040   int killed;
00041   struct file *ofile[NOFILE];
00042   struct inode *cwd;
00043   char name[16];
00044 };
```

```
// Saved registers for kernel context switches.
struct context {
  uint64 ra;
  uint64 sp;

  // callee-saved
  uint64 s0;
  uint64 s1;
  uint64 s2;
  uint64 s3;
  uint64 s4;
  uint64 s5;
  uint64 s6;
  uint64 s7;
  uint64 s8;
  uint64 s9;
  uint64 s10;
  uint64 s11;
};
```

xv6: swtch()

- Kernel/swtch.S

```
swtch:
        sd ra, 0(a0)
        sd sp, 8(a0)
        sd s0, 16(a0)
        sd s1, 24(a0)
        sd s2, 32(a0)
        sd s3, 40(a0)
        sd s4, 48(a0)
        sd s5, 56(a0)
        sd s6, 64(a0)
        sd s7, 72(a0)
        sd s8, 80(a0)
        sd s9, 88(a0)
        sd s10, 96(a0)
        sd s11, 104(a0)

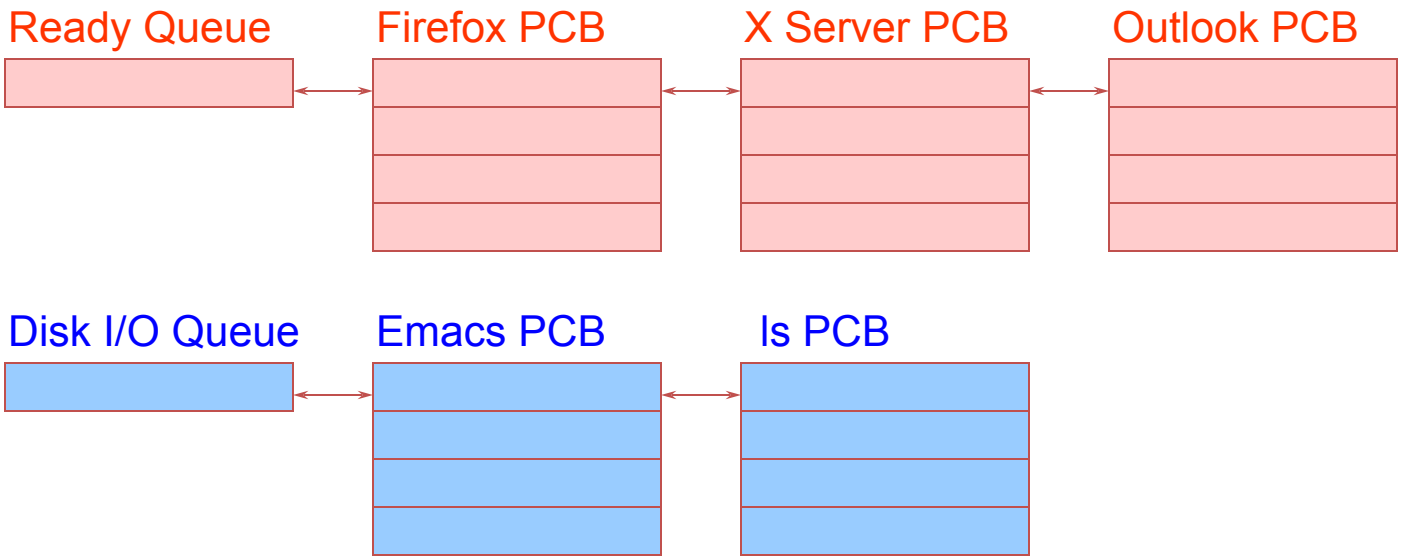
        ld ra, 0(a1)
        ld sp, 8(a1)
        ld s0, 16(a1)
        ld s1, 24(a1)
        ld s2, 32(a1)
        ld s3, 40(a1)
        ld s4, 48(a1)
        ld s5, 56(a1)
        ld s6, 64(a1)
        ld s7, 72(a1)
        ld s8, 80(a1)
        ld s9, 88(a1)
        ld s10, 96(a1)
        ld s11, 104(a1)

        ret
```

How does the OS track process states?

- The OS maintains a collection of queues that represent the state of all processes in the system
- Typically, the OS maintains at least one queue for each state
  - Ready, waiting, etc.
- Each process (namely PCB) is queued on a state queue according to its current state
- As a process changes state, its PCB is unlinked from one queue and linked into another

State Queues



Console Queue

Sleep Queue

.

.

.

There may be many wait queues, one for each type of wait (disk, console, timer, network, etc.)

Process System Call APIs

- Process creation: how to create a new process?
- Process termination: how to terminate and clean up a process
- Coordination between processes
  - wait, waitpid, signal, inter-process communication, synchronization
- Other
  - e.g., set quotas or priorities, examine usage, …

Process Creation

- A process is created by another process
  - Parent is creator, child is created (Unix: ps “PPID” field)
  - What creates the first process (Unix: init (PID 1))?

- After creating a child, the parent may either wait for it to finish its task or continue in parallel (or both)

Process Creation: Windows

- The system call on Windows for creating a process is called, surprisingly enough, CreateProcess:

  `BOOL CreateProcess(char *prog, char *args)` (simplified)

- CreateProcess
  - Creates and initializes a new PCB
  - Creates and initializes a new address space
  - Loads the program specified by "prog" into the address space
  - Copies "args" into memory allocated in address space
  - Initializes the saved hardware context to start execution at main (or wherever specified in the file)
  - Places the PCB on the ready queue

Process Creation: Unix

- In Unix, processes are created using fork(): `int fork()`
- **fork()**
  - Creates and initializes a new PCB
  - Creates a new address space
  - Initializes the address space with a **copy** of the entire contents of the address space of the parent
    - Child has the same memory and registers: same SP & PC
  - Initializes the kernel resources to point to the resources used by parent (e.g., open files)
  - Places the PCB on the ready queue
- fork() returns **twice**
  - Returns the child's PID to the parent, "0" to the child
  - Child and parent resume execution from this point: they have same PC

fork()

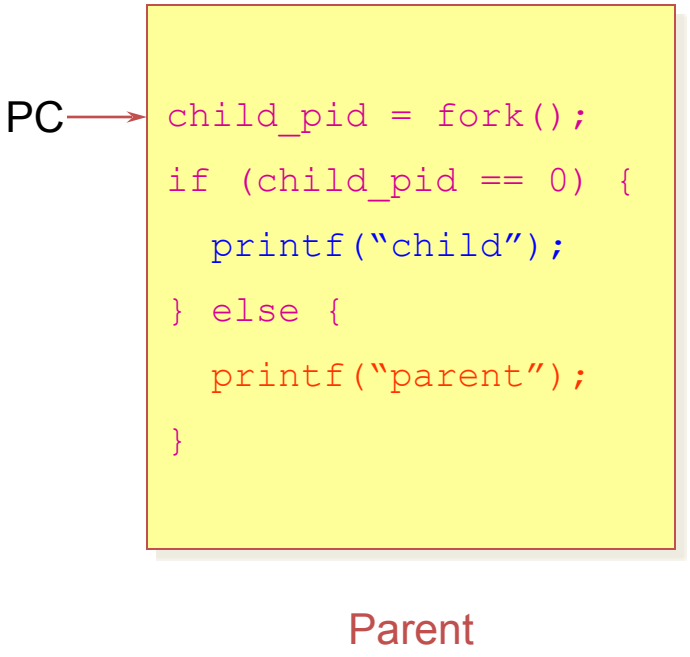
```
int main(int argc, char *argv[])
{
   char *name = argv[0];
   int child_pid = fork();
   if (child_pid == 0) {
        printf("Child of %s is %d\n", name, getpid());
        return 0;
   } else {
        printf("My child is %d\n", child_pid);
        return 0;
   }
}
```

What does this program print?

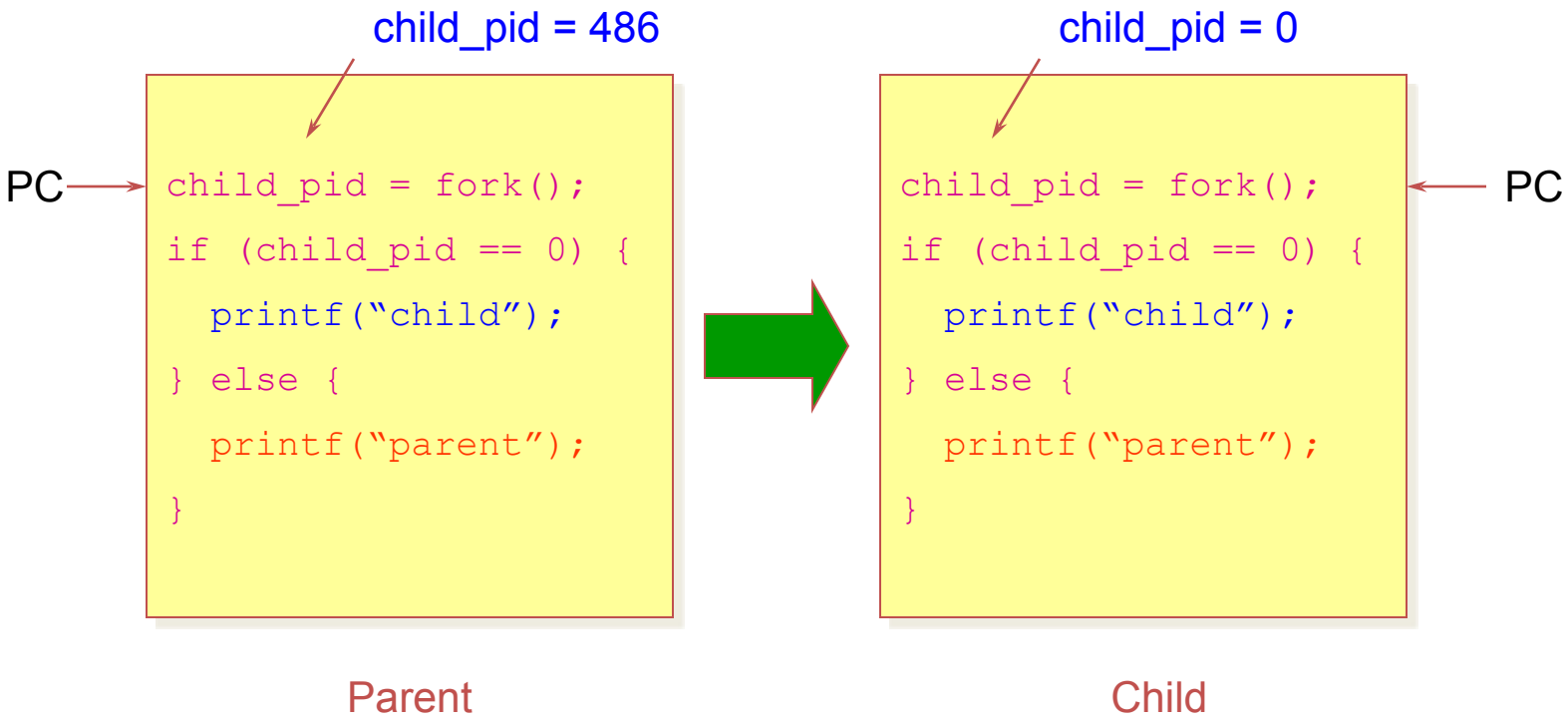
Example Output

```
[well ~]$ gcc t.c
[well ~]$ ./a.out
My child is 486
Child of a.out is 486
```

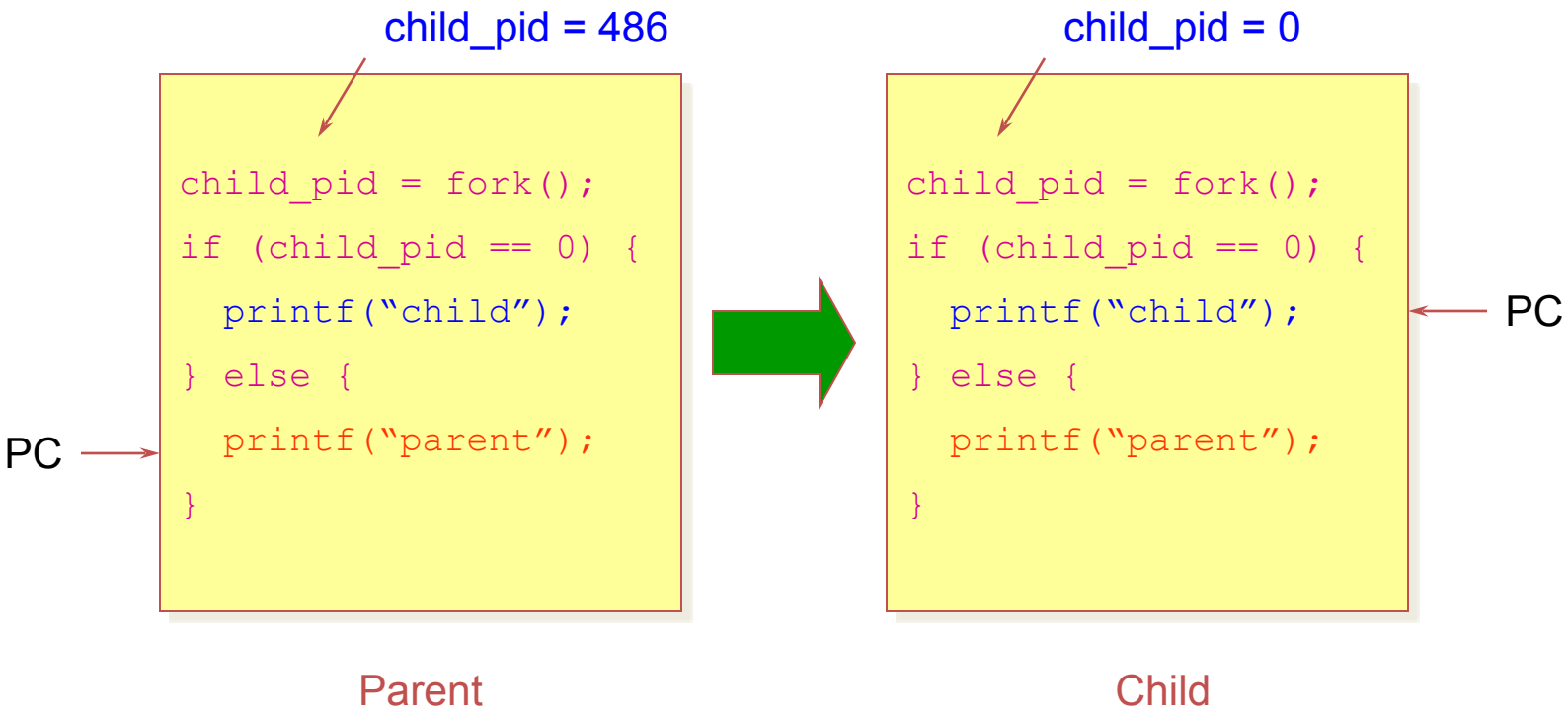
Duplicating Address Spaces



Duplicating Address Spaces



Divergence


child_pid = 486
child_pid = fork();
if (child_pid == 0) {
printf("child");
} else {
printf("parent");
}
PC
Parent
child_pid = 0
child_pid = fork();
if (child_pid == 0) {
printf("child");
} else {
printf("parent");
}
PC
Child

Example Continued

```
[well ~]$ gcc t.c
[well ~]$ ./a.out
My child is 486
Child of a.out is 486
[well ~]$ ./a.out
Child of a.out is 498
My child is 498
```

Why is the output in a different order?

Why fork()?

- Simple approach to enable concurrent execution
- Useful when the child…
  - Is cooperating with the parent
  - Relies upon the parent's data to accomplish its task

- Example: Web server

```
while (1) {
    int sock = accept();
    if ((child_pid = fork()) == 0) {
            Handle client request
    } else {
            Close socket
    }
}
```

Process Creation: Unix (2)

- Wait a second. How do we actually start a new program?

```
int exec(char *prog, char *argv[])
```

- **exec()**
  - Stops the current process
  - Loads the program "prog" into the process' address space
  - Initializes hardware context and args for the new program
  - Places the PCB onto the ready queue
  - Note: It **does not** create a new process

- What does it mean for exec to return?

Process Termination

- All good processes must come to an end. But how?
  - Unix: exit(int status), Windows: ExitProcess(int status)
- Essentially, free resources and terminate
  - Terminate all threads (next lecture)
  - Close open files, network connections
  - Allocated memory (and VM pages out on disk)
  - Remove PCB from kernel data structures, delete
- Note that a process does not need to clean up itself
  - OS will handle this on its behalf

wait() a second…

- Often it is convenient to pause until a child process has finished
  - Think of executing commands in a shell
- Use wait() (WaitForSingleObject)
  - Suspends the current process until a child process ends
  - Returns the process ID of the terminated child
  - waitpid() suspends until the specified child process ends
- Unix: Every process must be reaped by a parent
  - Basically, reading its child's exit status using wait()
  - What happens if a parent process exits before a child?
    - Orphan process; adopted by init (PID 1)
  - "zombie" process: undead; a process that has died but has not yet been reaped by the parent

Unix Shells

```
while (1) {
   char *cmd = read_command();
   int child_pid = fork();
   if (child_pid == 0) {
        Manipulate STDIN/OUT/ERR file descriptors for pipes, redirection,
   etc.
        exec(cmd);
        panic("exec failed");
   } else {
        if (!(run_in_background))
              waitpid(child_pid);
   }
}
```

Child (new prog)

Parent (shell)

Discussion Questions

Process A creates Process B using fork(). After creation, which of the following are true about what A and B share or do not share (in a traditional Unix-like OS)?

A. They share the same stack memory.
B. They have separate copies of CPU registers, including the program counter.
C. B inherits from A the open file descriptors.
D. They share the same heap memory by default.
E. Both parent and child continue execution from the same instruction
F. The return value of fork() distinguishes parent from child

Discussion Questions

Why is the OS abstraction of processes both powerful and expensive to manage?

A. Each process has its own memory space, providing strong isolation
B. Context switching between processes adds performance overhead
C. Processes require the OS to allocate and manage separate kernel resources for each instance
D. Processes inherently share memory by default, which can lead to data corruption

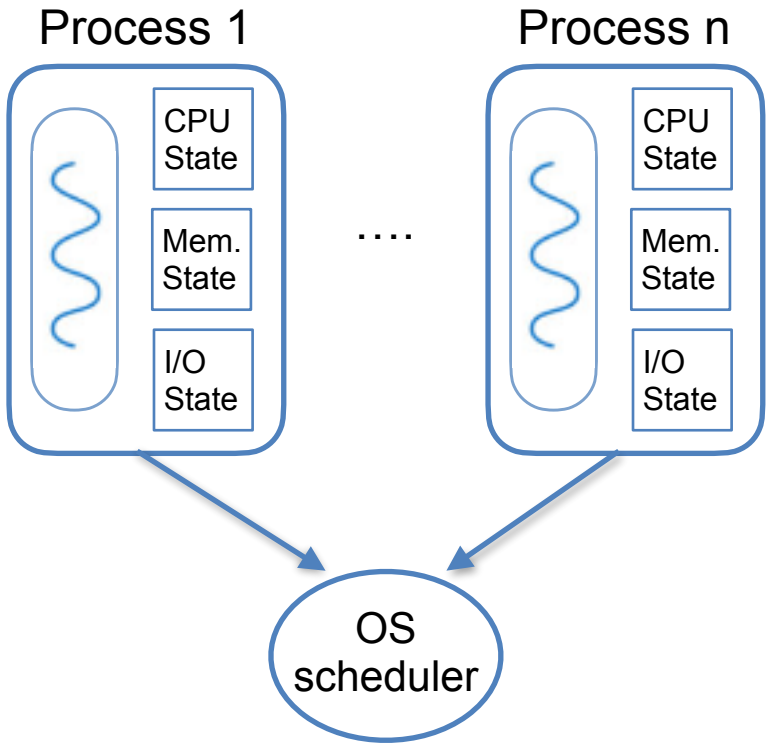
Some issues with processes

- Creating a new process is costly because of new address space and data structures that must be allocated and initialized
  - Recall struct proc in xv6

- Communicating between processes is costly because most communication goes through the OS
  - Each process has a **separate address space**
  - Inter Process Communication (IPC) – we will discuss later
  - Overhead of system calls and copying data

Process-associated overheads

- Context switch: **high**
  - CPU state: **low**
  - Memory / IO state: **high**
- Process creation: **high**
- Protection
  - CPU: **yes**
  - Memory / IO: **yes**
- Sharing overhead: **high**
  - At least one context switch



Parallel Programs

- Recall our Web server example that forks off copies of itself to handle multiple simultaneous requests

- To execute these programs, we need to
  - Create several processes that execute in parallel
  - Cause each to map to the same address space to share data
    - They are all part of the same computation
  - Have the OS schedule these processes in parallel

- This could be very inefficient
  - Space: PCB, page tables, etc.
  - Time: create data structures, fork and copy address space, etc.

Rethinking Processes

- What is similar in these cooperating processes?
  - They all share the same code and data (address space)
  - They all share the same privileges
  - They all share the same resources (files, sockets, etc.)
- Why don't they share resources?
  - While each keeps its own execution state: PC, SP, and registers
- Key idea: Separate *resources* from *execution state*

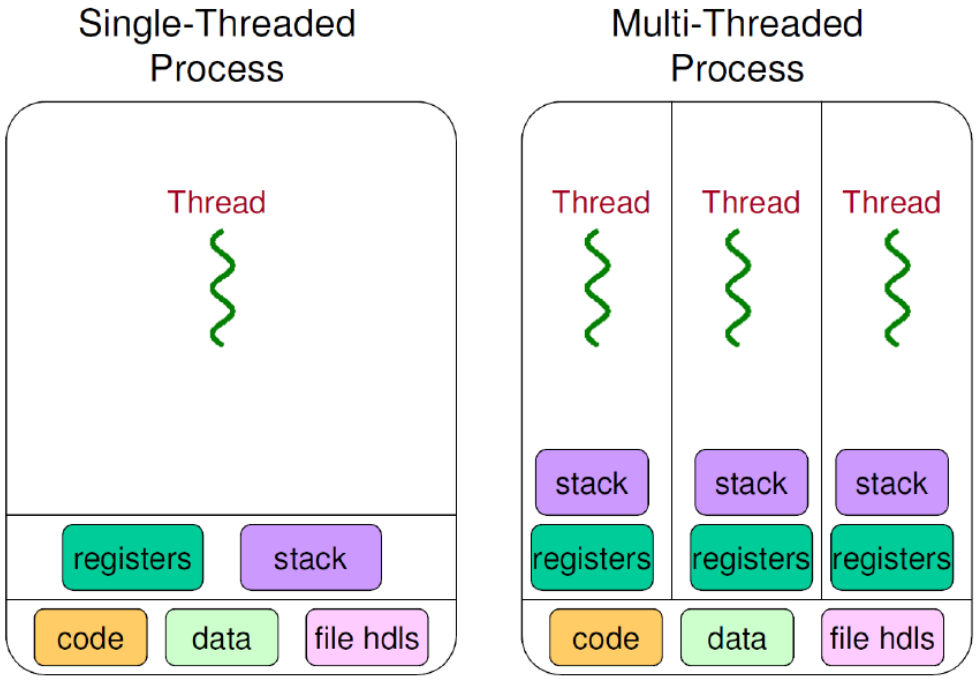
Threads

- Separate *execution* and *resource container* roles
  - The thread defines a *sequential execution stream* within a process (PC, SP, registers)
  - The process defines the address space, resources, and general process attributes (everything but threads)

Threads

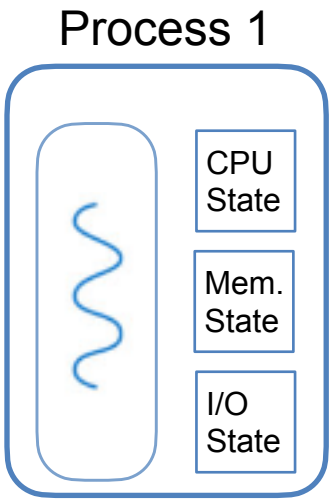
- Separate *execution* and *resource container* roles
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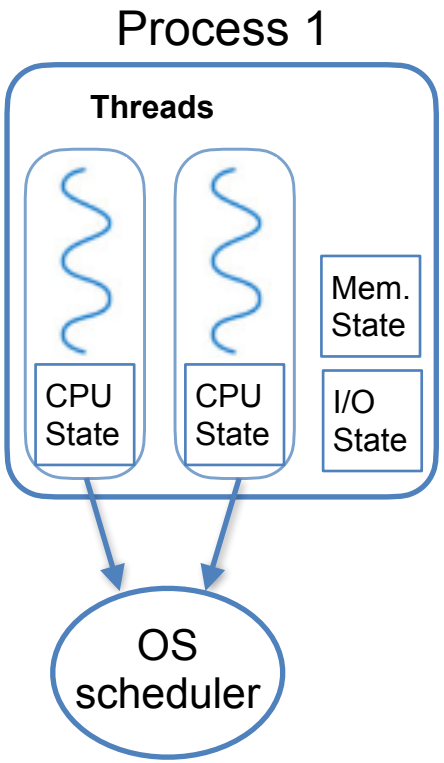
- Threads become the unit of scheduling
  - Processes are now the containers in which threads execute
  - Processes become static, threads are dynamic schedulable entities



Thread-associated overheads

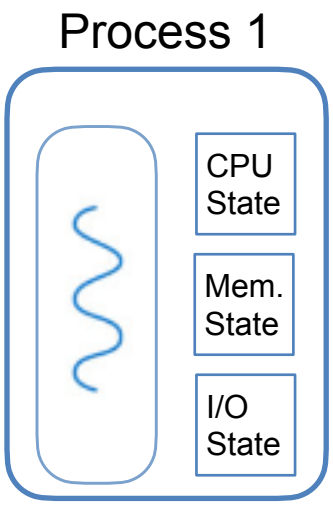
- Context switch: **low**
  - Only CPU state
  - No more memory / IO state
- Thread creation: **low**
- Protection
  - CPU: **yes**
  - Memory / IO: **no**
- Sharing overhead: **low**
  - Due to low thread switch

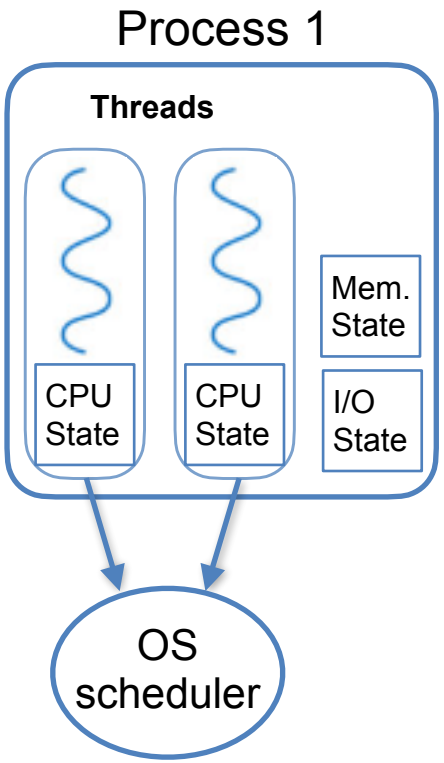




Thread-associated overheads

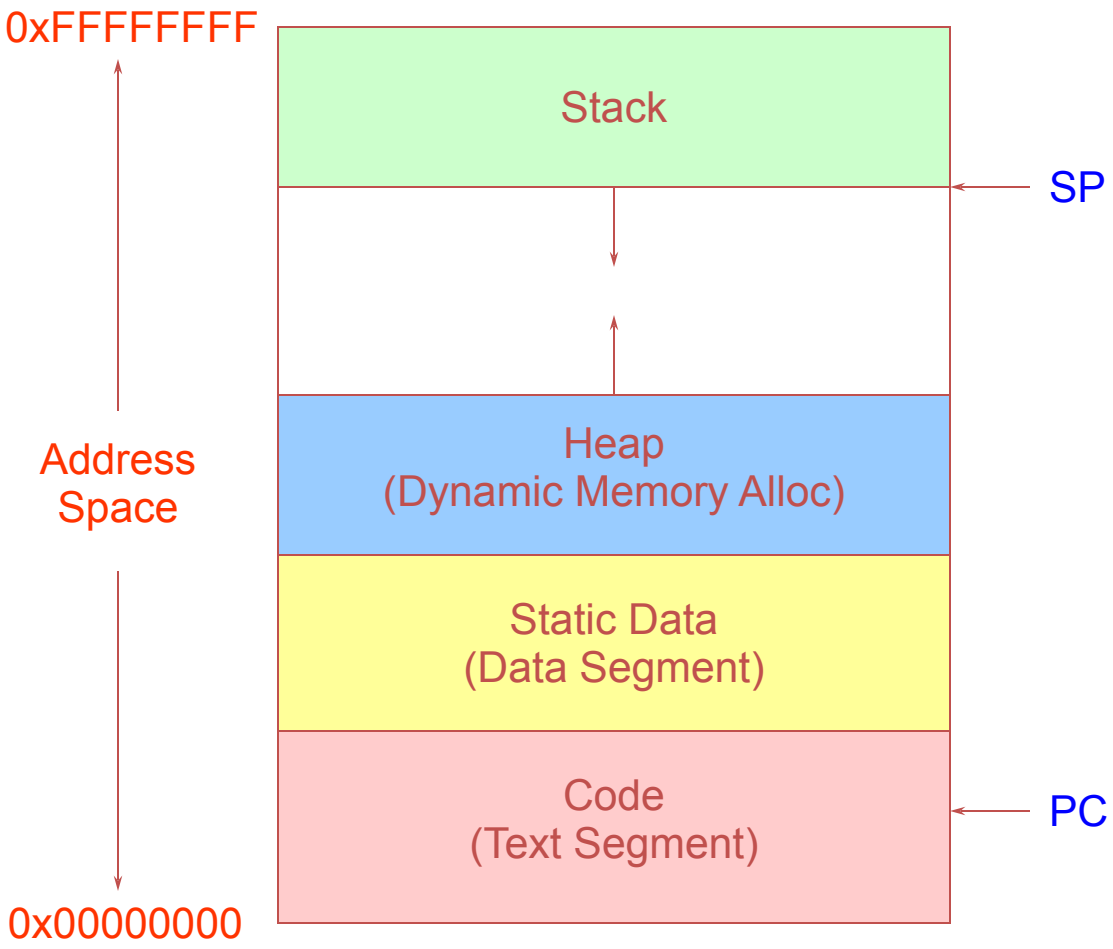
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- Sharing overhead: **low**
  - Due to low thread switch



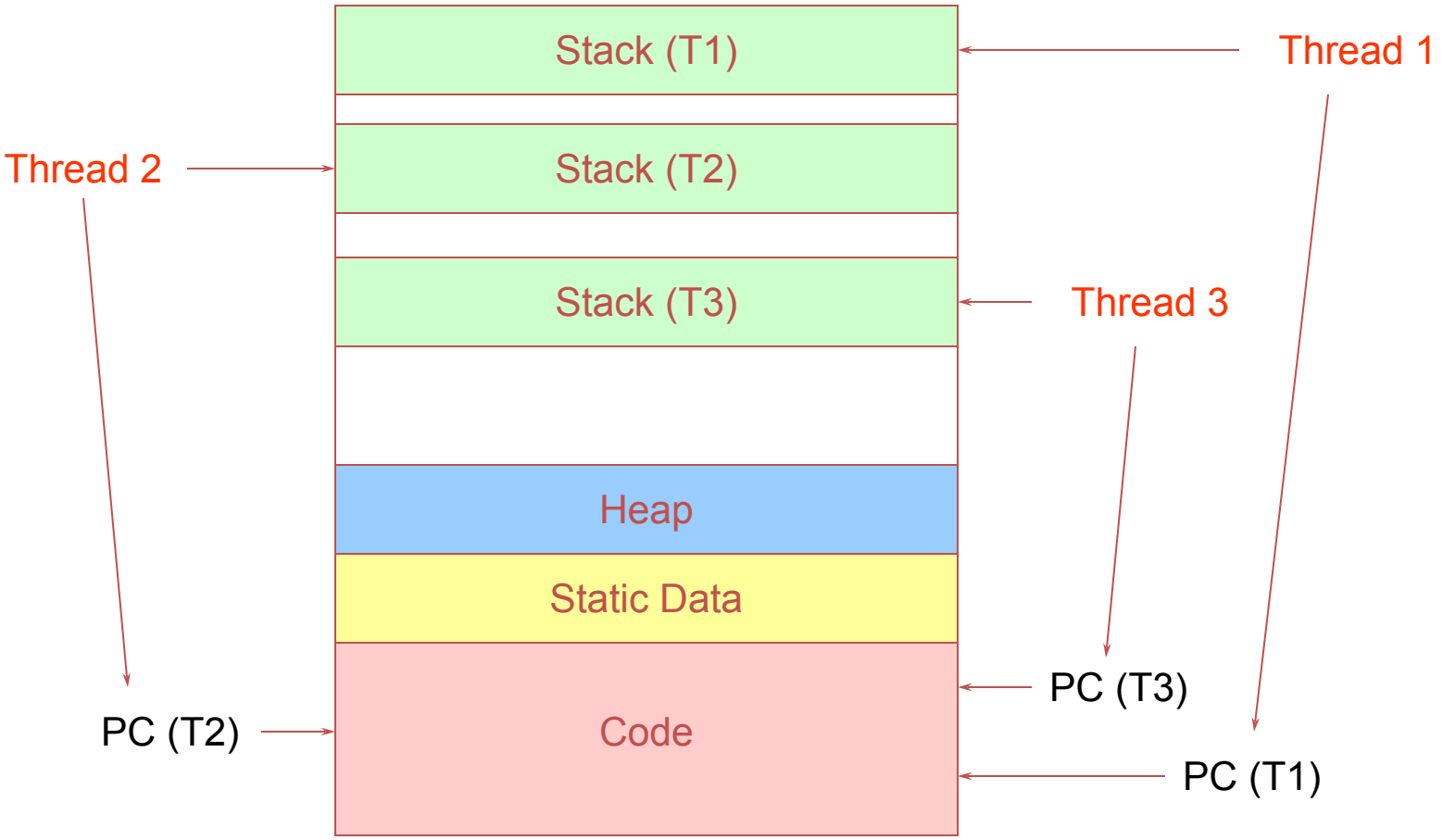


Context switching mainly depends on the process or thread's hunger for memory (i.e., working set size)

Recap: Process Address Space



Threads in a Process



Threads: Concurrent Servers

- Using fork() to create new processes to handle requests in parallel is overkill for such a simple task

- Recall our forking Web server:

```
while (1) {
    int sock = accept();
    if ((child_pid = fork()) == 0) {
            Handle client request
            Close socket and exit
    } else {
            Close socket
    }
}
```

Threads: Concurrent Servers

- Instead, we can create a new thread for each request

```
web_server() {
    while (1) {
        int sock = accept();
        thread_fork(handle_request, sock);
    }
}

handle_request(int sock) {
    Process request
    close(sock);
}
```

Sample Thread Interface

- thread_fork(procedure_t)
  - Create a new thread of control
  - Also thread_create(), thread_setstate()
- thread_stop()
  - Stop the calling thread; also thread_block
- thread_start(thread_t)
  - Start the given thread
- thread_yield()
  - Voluntarily give up the processor
- thread_exit()
  - Terminate the calling thread; also thread_destroy

Discussion Questions

Which of the following best explains why context switching between threads is generally less costly than context switching between processes?

A. Threads do not require saving and restoring registers during a context switch.
B. Threads share the same memory space, avoiding memory-related operations during a switch.
C. Process switches use preemptive scheduling, which takes more time than cooperative scheduling.
D. The OS kernel caches thread states in hardware, while process states are stored in main memory.

Discussion Questions

Which of the following are true about threads in a multithreaded program on a multicore Linux system?

A. Threads can run in parallel on multiple CPU cores
B. Threads always run on the same core as the main thread for cache locality
C. Shared memory access by threads can lead to race conditions
D. Threads are isolated from each other and cannot share data

Implementing threads

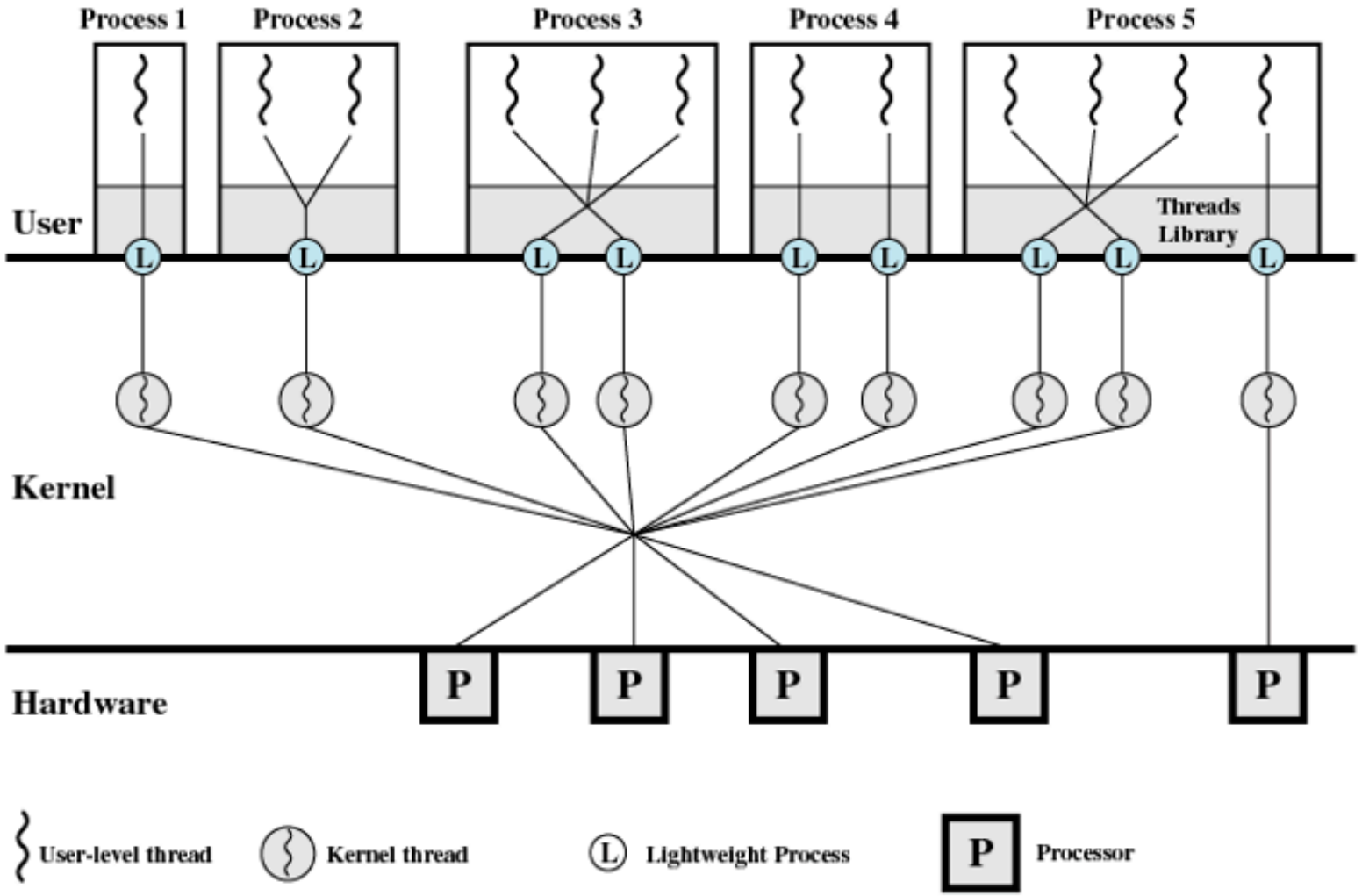
- Kernel-Level Threads
  - Their management is done in the kernel
  - The OS schedules all of the threads in the system
    - Requires a full thread control block (TCB) for each thread
  - Don't have to separate from processes

- OS-managed threads are called kernel-level threads or lightweight processes
  - Windows: threads
  - Solaris: lightweight processes (LWP)
  - POSIX Threads (pthreads): PTHREAD_SCOPE_SYSTEM

Alternative: User-Level Threads

- Implement and manage threads entirely at user level
  - Kernel knows nothing about user-level threads
- ULTs are small and fast
  - A thread is simply represented by a PC, registers, stack, and small thread control block (TCB)
  - Creating a new thread, switching between threads, and synchronizing threads are done via user-level procedure call
    - No kernel involvement; No mode switching
  - User-level thread operations can be 100x faster than kernel threads
  - ULTs are required to use non-blocking system calls. (Why?)
  - pthreads: PTHREAD_SCOPE_PROCESS

KLT and ULT combined



(Operating Systems, Stallings)

Figure 4.15 Solaris Multithreaded Architecture Example

Summary KLT vs. ULT

- Kernel-level threads
  - Integrated with OS (informed scheduling)
  - Slow to create, manipulate, synchronize
- User-level threads
  - Fast to create, manipulate, synchronize
  - Not integrated with OS (uninformed scheduling)
- Understanding the differences between kernel and user-level threads is important
  - For programming (correctness, performance)
  - For fundamental systems knowledge

Discussion Questions

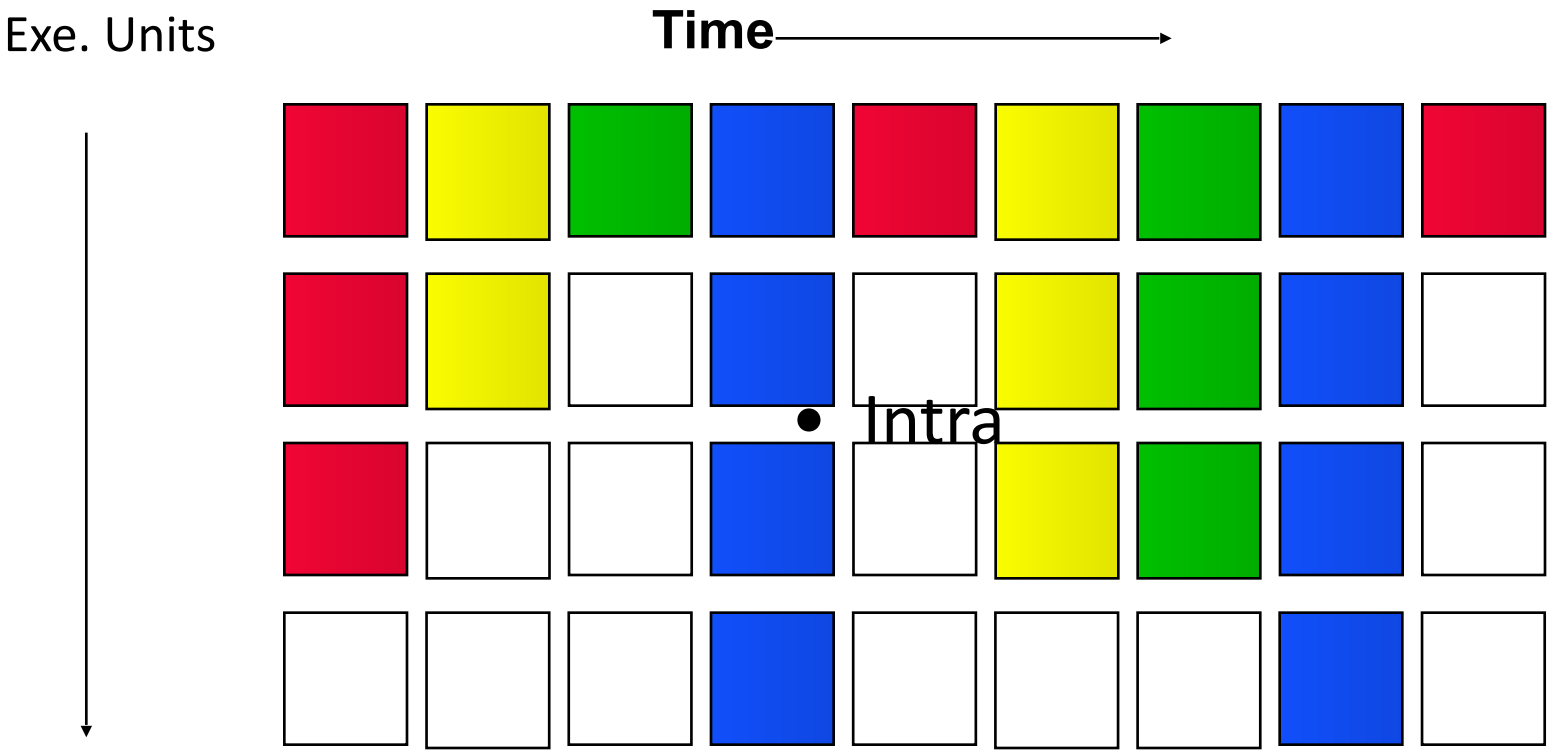
What challenges must developers consider when using threads in performance-sensitive applications?

A. Proper use of synchronization to avoid race conditions
B. Deadlocks caused by improper locking strategies
C. Each thread must be run on a separate core, limiting scalability
D. Shared data structures require careful coordination

Simultaneous multithreading & hyperthreading

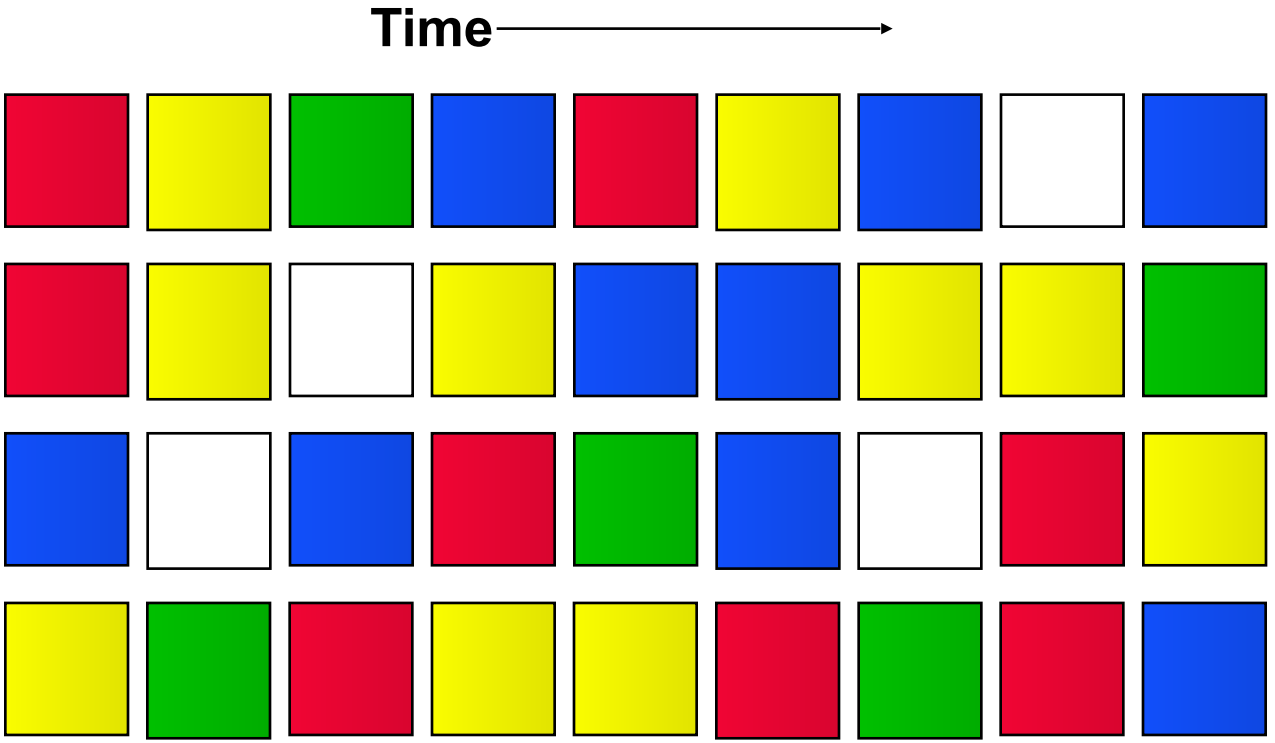
- Traditional multithreading can start execution of instructions from **only** a single thread at a given cycle
  - Low utilization if not enough instructions from a thread to dispatch in one cycle
  - Bad for machines with multiple execution units (i.e., superscalar architecture)
- Idea: dispatch instructions from multiple threads in the same cycle to keep multiple execution units utilized
  - Hirata et al., "An elementary processor architecture with simultaneous instruction issuing from multiple threads," ISCA 1992
  - Tullsen et al., "Simultaneous Multithreading: maximizing on-chip parallelism," ISCA 1995

Multithreading



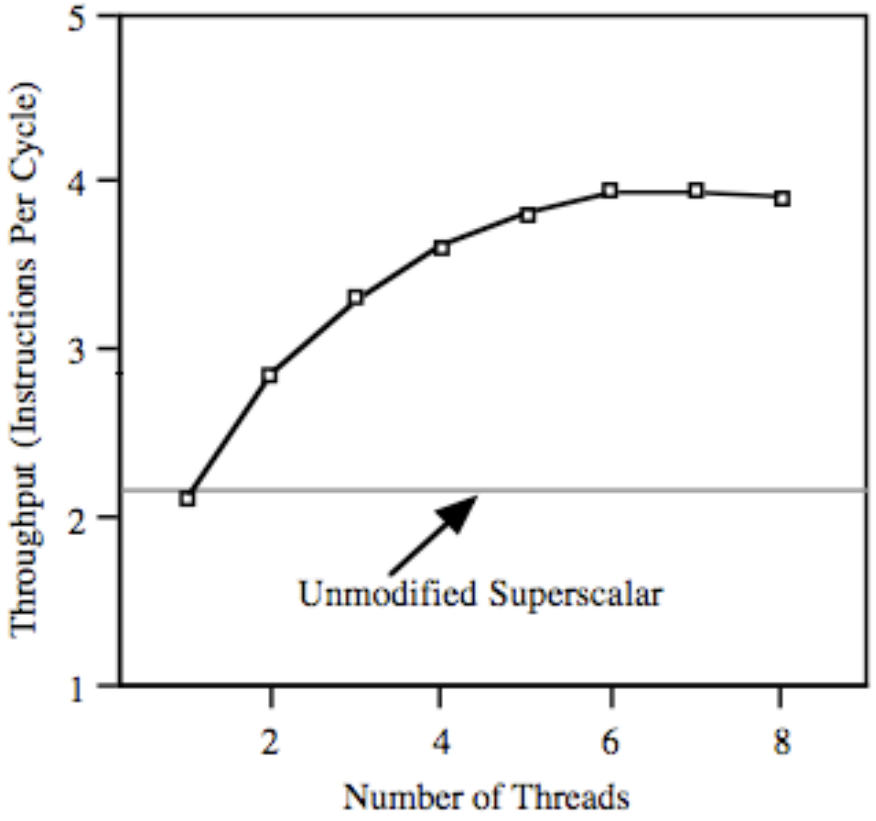
- Intra-thread dependencies
- Single thread performance suffers

SMT



- Utilize functional units with independent operations from multiple threads

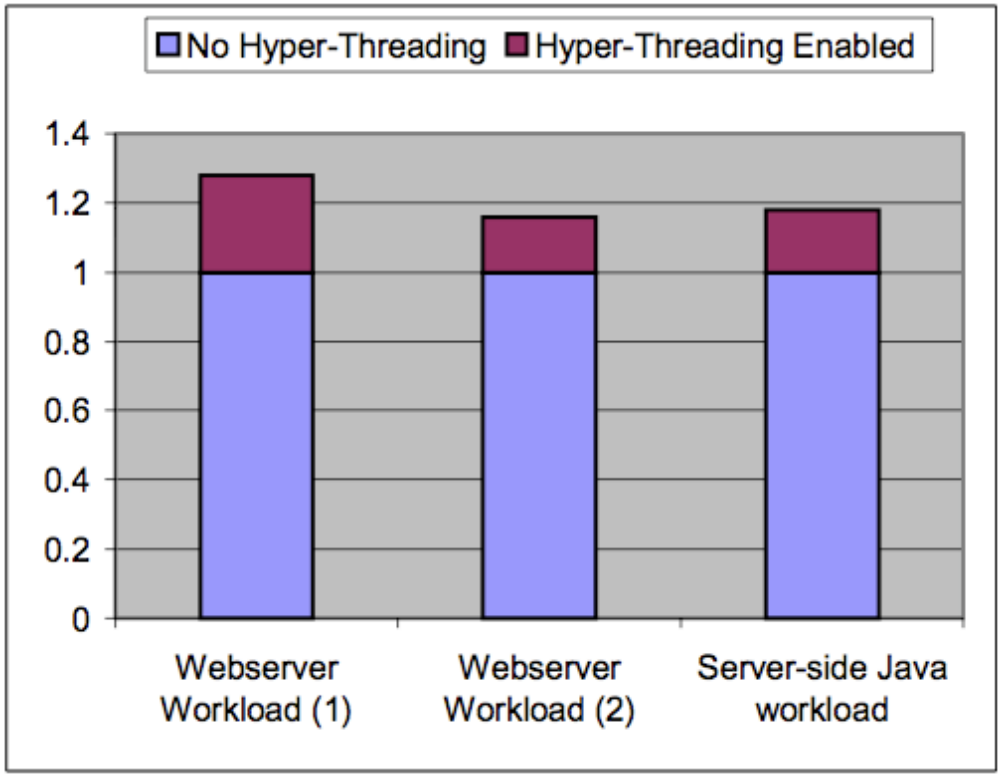
SMT scalability



Commercial SMT implementations

- Intel Pentium 4 (Hyperthreading)
- IBM POWER5
- Intel Nehalem

Intel Pentium 4 HT



- Marr et al., "Hyper-threading technology architecture and microarchitecture," Intel technology journal, 2002

Summary: why processes & threads?

- Goals
  - Multiprogramming: run multiple applications concurrently
  - Protection: don't want a bad application to crash system
- Solution
  - **Process**: given process/program illusion that it owns hardware resources
- Challenges
  - Process creation & switching is expensive
  - Concurrency within a same application
- Solution
  - **Thread**: decouple allocation and execution
  - Multiple threads within same process with less context switch overhead (only CPU state)